

Reading the Bode Plot

Ride and grip in the frequency domain — what the curves mean, and what "good" looks like for different cars. MICKSUS · Sweeps > Ride/grip frequency response

1 · Where the curves come from

The suspension is reduced to the standard **quarter car**: a sprung corner mass m_s on the wheel-rate spring k_s and damper c , sitting on an unsprung mass m_u that rests on the tire spring k_t . The road pushes the bottom; everything else follows.

Newton on the two masses, with z measured upward:

$$m_s z''_s = -k_s(z_s - z_u) - c(z'_s - z'_u)$$

$$m_u z''_u = +k_s(z_s - z_u) + c(z'_s - z'_u) - k_t(z_u - z_r)$$

Substituting $z = Z \cdot e^{st}$ turns the differential equations into a 2x2 algebraic system in $s = j\omega$. Inverting it gives every curve on the screen as a ratio to the road input Z_r — that ratio, plotted against frequency, is the Bode magnitude plot. The tool sweeps 0.2–30 Hz on a log axis.

Two natural frequencies fall out. Approximately:

$$f_{\text{body}} \approx (1/2\pi) \cdot \sqrt{(k_{\text{ride}}/m_s)} \text{ with } k_{\text{ride}} = k_s k_t / (k_s + k_t) \quad f_{\text{hop}} \approx (1/2\pi) \cdot \sqrt{((k_s + k_t)/m_u)}$$

MICKSUS does not use those approximations for the numbers it reports — it solves the undamped eigenproblem, which is exact and independent of damping. (Peak-picking off a damped curve is unreliable: near $\zeta = 1$ the peaks flatten, shift, and can disappear.)

2 · The four channels

Curve	Reads as	Who cares
Body acceleration (g per in of road)	Harshness. What fatigues the driver and shakes the electronics.	Everyone; dominant for endurance events.
Tire deflection (in per in)	Dynamic tire load variation. Large = the tire is unloading and about to skip. This is the grip channel.	Anyone who needs the tire to keep working on a rough surface.
Body motion (in per in)	Isolation. Below 1.0 the body moves less than the road.	Ride quality, camera/aero platform.
Suspension travel (in per in)	Rattle space consumed. Tells you where you will run out of stroke.	Long-travel cars, and anyone bottoming.

3 · Four rules for reading it

- **Below the body mode the curve sits at 1.0.** The whole car simply rides up and over the bump. Nothing you do to the suspension changes this — it is geometry, not tuning.
- **Isolation does not begin at the resonance — it begins at $\sqrt{2} \times f_{\text{body}}$.** Between f_{body} and $1.41 \cdot f_{\text{body}}$ transmissibility is still above 1, i.e. you are *amplifying* the road. This crossover is exact and independent of damping.
- **Above $\sqrt{2} \times f_{\text{body}}$, more damping makes isolation WORSE.** This is the rule that surprises people. Damping buys you a lower resonant peak and pays for it with worse high-frequency isolation. There is no setting that wins everywhere — pick which end matters.
- **The second peak is wheel hop**, the unsprung mass bouncing on the tire spring. Look at it on the *tire-deflection* curve, not the body curve: that is where losing the contact patch shows up.

4 · The trade you are actually making

Adding damping lowers the body-acceleration peak (better ride at resonance) but raises high-frequency transmission (worse over sharp, small bumps), and its effect on tire load is non-monotonic — too little and the wheel hops, too much and the wheel cannot follow the ground at all. Springs move the resonances; dampers only change the peaks' height and the roll-off. Unsprung mass is the free lunch: reducing it raises f_{hop} away from the excitation and shrinks the tire-load peak with no penalty anywhere.

What the y-axis should read

The frequency targets tell you *where* the peaks belong. This tells you *how tall* they should be. The key is that **three of the four channels are dimensionless ratios per inch of road**, so their targets are the same for a Baja car, an RC car and a Formula car — they are set almost entirely by the damping ratio. Only body acceleration is dimensional, and it is *not* comparable between classes.

5 · Damping sets the peak heights

Every peak height below comes from the quarter-car model itself, swept over damping ratio. Read down to your ζ and that is what the curves should read at their peaks:

ζ (bump)	Peak body motion (in/in)	Peak tire deflection (in/in)	Peak susp. travel (in/in)	Verdict
0.15	5.2	3.2	4.0	Far too little — floaty and skipping
0.25	3.3	2.2	2.5	Soft end of usable; off-road grip bias
0.35	2.5	1.8	1.9	Good all-round target
0.45	2.1	1.6	1.6	Milliken's road-holding optimum
0.55	1.9	1.5	1.4	Firm; platform control bias
0.70	1.8	1.4	1.3	Practical ceiling
0.90	1.7	1.5 ↑	1.3	Past useful — note tire deflection gets WORSE again

*That last row is the important one. Body motion keeps flattening as you add damping, but **tire deflection turns around and gets worse** past about $\zeta = 0.7$: the wheel can no longer move fast enough to follow the ground. That is the numerical fingerprint of over-damping, and it is why there is a ceiling rather than 'more is better'.*

6 · What each curve should read, and where

Channel	At DC (far left)	At the body mode	At wheel hop	Far right
Body motion (in/in)	exactly 1.0 — always; if not, something is wrong	the peak — 2–3 typical	0.2–0.3	→ 0
Tire deflection (in/in)	0	0.2–0.6	the peak — 1.5–2.0	→ 1.0
Suspension travel (in/in)	0	1.2–1.7	1.3–1.6	→ 0
Body accel (g/in)	0	1–2 g/in for a 2 Hz car	usually the peak — scales as f^2	→ 0

*The two hard anchors are worth memorising: **body motion starts at exactly 1.0** (the car rides over the bump) and **tire deflection ends at exactly 1.0** (the wheel stops following, so the tire swallows the whole input). If either asymptote is wrong, the model inputs are wrong.*

7 · Turning a ratio into a physical limit

A ratio only becomes a decision when you multiply it by a real bump height. Three conversions do all the work:

- **Tire lift-off.** dynamic load = tire-deflection $\times k_t \times$ bump. The tire leaves the ground when that reaches the static corner load. Worked: the Baja corner peaks at 1.8 in/in with $k_t = 251$ lb/in $\rightarrow$ 443 lb per inch of road, so only **0.29 in** of sustained input at the hop frequency fully unloads a 130 lb corner. The stiff-tired road car is worse — 1529 lb/in, so **0.10 in**. This is why periodic washboard is so destructive: it is the one input that actually sustains a resonance.
- **Bottoming.** travel needed = suspension-travel $\times$ bump. At 1.9 in/in a 3 in bump asks for 5.7 in of stroke; compare against what the car has before the bump stop.
- **Driver load.** g felt = body-accel $\times$ bump. This is the one that makes body acceleration comparable between classes — see below.

8 · Why body acceleration is not comparable between cars

Body acceleration is the only dimensional channel, and it scales as f^2 . The peak values from the model are wildly different — **Baja 2.5 g/in, road race 6.4 g/in, 1:10 RC 34 g/in** — which makes the RC car look thirteen times harsher than the Baja car. It is not. The bumps it meets are ten times smaller. Multiply through by a representative bump and the numbers converge: a Baja car over a 3 in rock sees ≈ 7.6 g; an RC car over a 0.3 in ripple sees ≈ 10 g. **Always multiply g/in by the bump size your car actually meets before comparing anything.**

What "good" looks like — three very different cars

	Baja SAE	1:10 scale RC	On-road race car
Body (ride) frequency	1.5 – 2.5 Hz (typ. ≈ 2)	5 – 8 Hz (Froude-scaled; see below)	1.5 – 2.5 Hz non-aero 3.0 – 5.0+ Hz with downforce
Wheel hop	9 – 12 Hz — low, because unsprung mass is heavy (a 15 lb tire) and the tire is soft	30 – 45 Hz — everything is light and stiff	12 – 18 Hz — stiff slick, light unsprung
Damping ratio (bump)	0.25 – 0.45. Lower end: the wheel must be free to follow rough ground	0.3 – 0.6, set by oil weight rather than a number	0.3 – 0.7. Higher for wheel control, lower for mechanical grip
Which curve rules	Tire deflection and suspension travel. You are bump-limited, not grip-limited	Body acceleration — landings and chassis settling dominate lap time	Tire deflection. Contact-patch load consistency is lap time; with aero, platform control
Peak y-values body motion / tire defl / body accel	$\approx 2.5 / 1.8 / 2.5$ g per in (at ζ 0.35)	$\approx 2.0 / 2.0 / 34$ g per in (at ζ 0.40 — high only because f is high; multiply by a 0.3 in ripple)	$\approx 1.8 / 1.9 / 6.4$ g per in (at ζ 0.45)
Classic failure	Wheel hop lands on washboard at trail speed → the tire skips and drive goes away	Over-sprung and under-damped → the car pogo off every bump and lands badly	Too much damping chasing platform control → mechanical grip vanishes on kerbs

Baja SAE — the rough-ground case

You are the softest and heaviest-unsprung of the three, which puts wheel hop unusually low (≈ 10 Hz). That matters because low-frequency hop is easy for terrain to excite: bumps spaced λ apart at speed V drive $f = V/\lambda$, so 2 ft washboard at only ~ 14 mph sits right on it. Watch the **tire-deflection** curve at 9–12 Hz and use the terrain table in the dialog to check whether that speed is one you actually drive. The single most effective fix is **less unsprung mass**, which moves hop up and shrinks the peak simultaneously.

Do not chase a low body-acceleration peak with damping. Off-road, the wheel needs to be free to move; over-damping is why a stiff-feeling car loses drive over whoops.

1:10 scale RC — why the numbers are $\sim 3\times$ higher

Gravity does not scale. Under **Froude scaling** (the standard similarity law when gravity drives the motion), if length scales by λ then time scales by $\sqrt{\lambda}$ and **frequency scales by $1/\sqrt{\lambda}$** . For 1:10 that is $\sqrt{10} \approx 3.2\times$. So a 2 Hz full-size ride frequency becomes ≈ 6.3 Hz, and a 12 Hz hop becomes ≈ 38 Hz. Speed scales the other way: 40 mph full-size corresponds to only ≈ 13 mph at 1:10.

Be careful with this. Real RC cars are documented not to achieve dynamic similarity — spring rates and resonant frequencies do not scale cleanly at 1:10, and RC cars are typically both stiffer and faster than true Froude scaling would require. Treat the scaled numbers as the target for a car meant to behave like a scaled race car, not as a description of a shelf buggy. Published RC setup guidance is empirical (spring colours, oil weights) rather than frequency-based.

On-road race car — the contact-patch case

The surface is smooth, so isolation matters far less and **tire load variation** is nearly the whole story: every pound of dynamic load fluctuation is grip you are not using. Non-aero cars sit at 1.5–2.5 Hz; downforce cars run 3–5+ Hz because aerodynamic platform control outranks mechanical compliance. Wheel hop moves up to 12–18 Hz on stiff slicks with light unsprung corners, excited by kerbs and surface texture rather than terrain.

*A useful cross-check for any car: the rear ride frequency is normally set **10–20% above the front** (Olley's flat-ride guidance), so that the rear catches up with the front after a bump and the car settles flat rather than pitching.*

Quick diagnostic checklist

- **Body and hop peaks both sharp and tall** → under-damped. Expect a floaty body and a skipping tire.
- **Peaks flat but high-frequency curve elevated** → over-damped. The car will feel harsh and lose grip over small sharp bumps. Check ζ : above ~ 0.7 in bump you are past the useful range.
- **Tire-deflection peak large at hop** → the contact patch is unloading. Reduce unsprung mass first, then soften the tire or add damping around the hop frequency.
- **A resonance sitting on a speed you drive a lot** (terrain table) → move the frequency, not the damping.

Sources. Ride frequency by class — M. Giaraffa, *Tech Tip: Springs & Dampers, Part One*, OptimumG (0.5–1.5 Hz passenger, 1.5–2.0 sedan racecar/moderate downforce, 3.0–5.0+ high downforce); Penske Racing Shocks, *Natural Frequency, Ride Frequency and CPM in Race Suspension* (desert 1.0–1.5, non-aero road race 1.5–2.5, stock/sprint 2.0–3.0). Damping ratios and the ~ 0.45 road-holding optimum — Milliken, *Race Car Vehicle Dynamics* (SAE). Wheel hop 5–20 Hz (typ. 10–15) and the $\sim 3:1$ rebound:bump convention — Gillespie, *Fundamentals of Vehicle Dynamics* (SAE). The $\sqrt{2}$ isolation crossover, and damping degrading transmissibility above it, are standard mechanical-vibration results. RC caveat — published scaled-vehicle similitude work reports that spring rates and resonant frequencies do not scale faithfully at 1:10.